

DESIGN, DEVELOPMENT, AND DEPLOYMENT OF A RAG CONVERSATIONAL ASSISTANT BASED ON THE MODEL CONTEXT PROTOCOL (MCP) FOR INTELLIGENT ACCESS TO DNSI BUSINESS DOCUMENTS VIA SHAREPOINT CERP

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General Context and Problem Statement

1. Digital transformation within large organizations such as the DNSI (Digitization and Information Systems Department) is accompanied by a massive accumulation of technical and business documentation, primarily stored on collaborative platforms like SharePoint CERP. In the absence of advanced semantic interrogation mechanisms, access to this informational asset relies essentially on manual consultation or basic search functionalities, which are insufficient to meet complex and contextual needs. This situation creates increasing difficulty in rapidly locating relevant and actionable information, slowing down operational and decision-making processes, and revealing a clear need for more intelligent and interactive knowledge access tools.

2. The advent of Large Language Models (LLMs) offers a new natural language interface for knowledge access. However, their direct integration into a corporate environment raises critical reliability and security challenges. Generic hallucinations inherent in generative models, coupled with their lack of knowledge regarding the organization's private data (data absent from their training corpus), mandate the adoption of hybrid architectures. The scientific challenge here is to design a system capable of interfacing these models with the private SharePoint corpus in a secure, standardized, and verifiable manner.

Literature Review and State of the Art

3. Recent scientific literature identifies the RAG (Retrieval-Augmented Generation) architecture as the reference solution for grounding LLM responses in an external knowledge base (Lewis et al., 2020). This approach combines a dense information retrieval system with a text generator, allowing for knowledge updates without costly model retraining. The use of vector indexes, such as Faiss, enables semantic similarity search significantly superior to lexical methods for open-ended questions.

4. However, integrating heterogeneous data sources (data silos like SharePoint) traditionally relies on fragile and hard-to-maintain ad-hoc connectors. The state of the art is evolving towards standardizing interfaces between AI assistants and data sources, embodied by the recent emergence of the *Model Context Protocol* (MCP). This protocol aims to normalize how context is exposed to models, promising increased interoperability and security compared to proprietary integration solutions. Studying this protocol within a real-world application framework constitutes a novel research axis.

Proposed System Description

5. This final year project proposes the design of a modular RAG architecture, based on the principles of the Model Context Protocol (MCP), to enable unified access to business documents hosted on SharePoint CERP. The system is designed to act as an intelligent intermediary, retrieving relevant context via semantic search mechanisms before submitting it to a high-performance language model (Mistral AI). The objective is to ensure that every generated response is sourced and traceable, meeting the rigorous standards of the DNSI.

6. The technical solution revolves around an advanced processing pipeline including secure document ingestion, segmentation, and vectorization. Particular attention is paid to authentication and rights management, integrating an SSO (Single Sign-On) mechanism to ensure the assistant respects the ACLs (Access Control Lists) defined in SharePoint. The developed prototype, internally named "AntiGravity", serves as a proof of concept to validate this standardized architectural approach against real production constraints (corporate proxy, latency, confidentiality).

Methodology and Architecture

7. The research methodology adopted is "Design Science Research," iterating between theoretical design and practical implementation. The first phase focuses on developing MCP-compatible connectors for the extraction and pre-processing of unstructured data (PDF, DOCX) from SharePoint. The logical architecture separates the context server (managing the Faiss index and documents) from the conversational client, fostering horizontal scalability and increased maintainability.

8. The current development phase implements the user interface and the security layer. The application, developed in Python (Streamlit), integrates re-ranking and query classification algorithms to optimize result relevance. Deployment is planned on containerized infrastructure, complying with enterprise security standards. The final system evaluation will rely on quantitative metrics (document retrieval precision) and qualitative metrics (user satisfaction, response fluency).

Conclusion and Scientific Contributions

9. This first design phase confirms the relevance of coupling the RAG architecture with a standardized MCP-like approach to solve enterprise information access problems. The developed theoretical model overcomes the limitations of rigid connectors, offering the flexibility necessary to evolve with the DNSI information system. Initial tests validate the system's ability to synthesize complex information from multiple SharePoint documents with a minimal hallucination rate.

10. Beyond the technical implementation, this work contributes to understanding the challenges of integrating Generative AI into traditional "Enterprise IT" ecosystems. It proposes a reference framework for the secure deployment of cognitive assistants, highlighting the importance of data sovereignty and protocol interoperability. Future developments will aim to extend protocol coverage to other business data sources, thereby strengthening the organization's collective intelligence.

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